

Comprehensive feature list for Angular 19 and Angular 20

Angular 19 – Feature List

Core Framework

- Stable **Signals API** (for local and shared state)
- Coexistence of **Signals + RxJS**
- Zone.js–based change detection (default)
- Improved change detection performance
- Better tree-shaking

Component & Architecture

- Standalone Components fully supported
- Standalone Directives and Pipes
- NgModule still supported (legacy-friendly)
- Functional providers and APIs
- Improved dependency injection ergonomics

State Management

- Signals (signal, computed, effect)
- RxJS still primary for async/state-heavy apps
- Automatic cleanup with effects
- Reduced need for OnDestroy

Routing

- Functional route guards and resolvers
- Lazy loading with standalone components
- Route-level providers
- Improved preloading strategies

Forms

- Reactive Forms improvements
- Better type safety in form controls
- Signals usable alongside forms (external state)

SSR & Hydration

- Stable Server-Side Rendering (SSR)
- Improved hydration over earlier versions
- Better SEO support
- Faster initial render than Angular 18

Tooling & CLI

- Faster builds with partial esbuild usage
- Improved HMR (Hot Module Replacement)
- Enhanced error messages
- Better Angular CLI schematics

Performance

- Reduced bundle sizes
- Faster runtime compared to Angular 18
- Improved memory management

Developer Experience

- Better debugging for signals
- Improved documentation
- Backward compatibility preserved

Angular 20 – Feature List

Core Framework

- **Signals-first architecture**
- **Zoneless change detection (recommended)**

- Explicit, predictable rendering model
- Further reduced bundle size
- Faster runtime and startup time

Component & Architecture

- Standalone APIs are the **default**
- NgModule considered legacy
- Signal-based @Input() and @Output()
- Functional-first API design
- Cleaner component lifecycle

State Management

- Signals used across:
 - Components
 - Services
 - Router
 - Inputs/Outputs
- Fewer subscriptions
- Minimal memory-leak risk
- Reduced RxJS dependency (not removed)

Routing

- Signal-aware Router
- Route params and data exposed as signals
- Faster navigation
- Smarter preloading
- Improved lazy loading

Forms

- Better integration between Signals and Forms

- Reduced reliance on valueChanges
- Easier synchronization between UI and state

SSR & Hydration

- Fine-grained hydration
- Faster Time to Interactive (TTI)
- Improved partial hydration
- Better performance for large applications

Tooling & CLI

- Full esbuild-based build pipeline
- Much faster build and rebuild times
- Smarter CLI defaults
- Improved diagnostics and warnings

Performance

- Zoneless execution reduces overhead
- Faster rendering cycles
- Lower memory usage
- More deterministic UI updates

Developer Experience

- Less boilerplate
- Cleaner mental model
- Easier debugging
- More explicit data flow
- Future-proof Angular architecture

Feature Comparison Snapshot

Category	Angular 19	Angular 20
Signals	Stable & optional	Core & dominant
Change Detection	Zone.js default	Zoneless recommended
RxJS Usage	Heavy	Reduced
Standalone APIs	Recommended	Default
SSR & Hydration	Improved	Advanced
Build Performance	Fast	Faster
Architecture	Transitional	Future-ready

Instructor Recommendation

- **Angular 19** is ideal for **stabilizing or maintaining existing enterprise applications**
- **Angular 20** is best for **new projects, high-performance apps, and long-term scalability**

If you want next:

- Feature-wise **code examples**
- **Angular 19 → 20 migration checklist**
- **Angular 20 project structure**
- **Interview questions based on Angular 19 & 20**

Tell me how you want to proceed.